

tion of $\mathbf{P}$ is left up to the user. Recent work has shown that $\mathbf{P}$ must be chosen carefully when one applies modularity maximization to correlation matrices. One possible choice is to set $\mathbf{P} = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix (MacMahon and Garlaschelli, 2015; Bazzi et al., 2014). This definition corresponds to a null model where time series are uncorrelated with one another. Thus, community detection amounts to placing as many positive correlation coefficients within each community as possible. To normalize Q so that it is bounded between 0 and 1, we scale $Q(t)$ by $1/2m$, where $2m = \sum_{ij} |W_{ij}|$.

It is worth noting that the modularity function used here differs from most standard approaches for dealing with signed networks. Traditionally, signed networks are sub-divided into two separate networks: one containing just positive connections:

$$W_{ij}^+ = \begin{cases} W_{ij}, & \text{if } W_{ij} > 0 \\ 0, & \text{otherwise} \end{cases}$$

and

$$W_{ij}^- = \begin{cases} -W_{ij}, & \text{if } W_{ij} < 0 \\ 0, & \text{otherwise} \end{cases}$$

Modularity functions are then defined for both the positive and negative components as: $Q^\pm = \sum_{ij} [W_{ij}^\pm - P_{ij}^\pm] \delta(g_i, g_j)$. The choice of how to combine the two components to obtain the total modularity is left up to the user, though a number of weighting schemes have been proposed (Gómez et al., 2009; Rubinov and Sporns, 2011). In general, the total modularity is of the form: $Q = c^+ Q^+ - c^- Q^-$, where $c^\pm$ are constants. Thus, modularity maximization for signed networks can be viewed as an attempt to maximize the modularity of positive connections while penalizing negative connections when they fall within